

LEHIGH VALLEY, PA, USA

WMO: 725170

Lat: 40.650N

Lon: 75.448W

Elev: 390

StdP: 14.49

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.1	(c) 12.5	(d) -6.2	(e) 4.0	(f) 10.5	(g) -1.6	(h) 5.1	(i) 15.4	(j) 24.7	(k) 30.3	(l) 22.3	(m) 28.2	(n) 8.4	(o) 330	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	91.2	74.1	88.5	72.7	86.0	71.4	76.7	86.8	75.3	84.1	73.9	82.2	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	127.0	81.2	72.4	121.8	80.0	71.2	116.6	78.9	40.2	86.8	38.9	84.3	37.6	82.4	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.9	(b) 18.6	(c) 16.6	(e) 2.5	(f) 95.3	(g) 6.0	(h) 3.2	(i) -1.8	(j) 97.6	(k) -5.3	(l) 99.5	(m) -8.6	(n) 101.3	(o) -12.9	(p) 103.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			2.5	95.3	6.0	3.2	-1.8	97.6	-5.3	99.5	-8.6	101.3	-12.9	103.6		
			1.2	79.4	5.6	1.7	-2.8	80.6	-6.1	81.6	-9.2	82.5	-13.3	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	29.2	31.6	40.0	51.4	61.2	70.2	74.7	72.9	66.0	54.4	43.3	34.5
	DBStd	17.70	10.09	9.29	9.22	8.35	7.66	6.06	5.13	5.05	6.75	7.93	8.37	8.63
	HDD50	2339	646	516	332	78	5	0	0	0	43	234	485	
	HDD65	5435	1108	934	774	415	170	22	1	69	342	652	944	
	CDD50	3282	3	2	23	122	353	605	767	710	479	179	33	6
	CDD65	902	0	0	1	8	53	177	304	249	98	12	0	0
	CDH74	8242	0	1	15	132	597	1594	2885	2126	794	96	2	0
	CDH80	2745	0	0	2	31	183	503	1098	696	218	14	0	0
(14) Wind	WSAvg	6.9	7.8	8.1	8.6	8.3	6.9	6.4	5.7	5.2	5.7	6.4	6.7	7.2
(15) Precipitation	PrecAvg	44.1	3.3	2.8	3.4	3.5	3.9	3.8	4.3	4.1	4.4	3.3	3.6	3.5
	PrecMax	56.9	8.4	5.4	6.7	7.9	10.6	8.6	10.4	9.4	11.6	13.1	9.7	7.9
	PrecMin	29.8	0.7	0.6	0.9	0.6	0.1	0.9	0.3	0.7	0.7	0.2	0.7	0.6
	PrecStd	7.0	1.8	1.1	1.4	1.7	2.1	1.8	2.4	2.2	2.8	2.2	1.9	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	64.0	75.5	84.3	89.1	92.2	95.9	93.3	90.7	81.6	70.7	64.3
		MCWB	59.3	55.1	60.6	64.5	70.8	74.3	75.3	75.4	73.2	68.1	61.7	58.4
	2%	DB	54.4	55.7	66.5	77.5	85.3	88.5	92.0	89.6	85.6	76.4	66.2	57.0
		MCWB	50.3	47.6	54.0	60.9	68.9	72.7	74.5	74.0	70.0	65.5	58.8	52.4
	5%	DB	47.4	49.7	60.4	72.1	80.9	85.5	89.0	86.5	81.9	72.4	61.9	51.7
		MCWB	43.1	42.4	50.2	58.2	66.3	70.9	73.2	72.1	69.0	63.2	55.4	47.9
	10%	DB	43.1	44.8	55.1	67.1	76.3	82.8	86.1	83.9	78.6	68.3	57.2	47.5
		MCWB	39.1	39.5	46.9	54.2	63.6	69.3	72.2	70.7	67.3	60.3	51.1	42.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	57.8	62.0	68.3	73.8	77.1	78.9	78.4	75.8	71.2	64.3	59.5
		MCDB	62.6	63.8	71.7	79.7	85.6	87.9	90.7	89.1	85.6	77.4	68.1	63.2
	2%	WB	51.0	48.6	56.4	63.8	70.7	75.0	77.0	76.3	73.4	68.1	60.4	53.7
		MCDB	53.4	54.5	63.9	74.3	82.0	84.6	87.8	85.1	80.4	74.4	65.1	56.6
	5%	WB	44.1	43.6	51.9	60.0	68.4	73.3	75.4	74.8	71.6	65.0	56.3	48.4
		MCDB	47.0	48.3	58.7	69.2	77.8	82.3	84.8	82.5	78.1	70.6	60.4	51.2
	10%	WB	39.4	39.5	47.2	56.6	65.9	71.5	74.0	73.2	69.6	61.7	52.2	43.5
		MCDB	42.4	44.6	54.3	65.0	74.0	79.5	82.7	80.5	75.9	67.3	56.4	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	16.5	18.5	21.4	20.8	19.7	19.9	19.6	20.4	19.8	17.9	14.5
		MCDBR	20.4	23.7	27.2	29.4	27.1	23.0	23.1	22.4	24.3	24.7	23.6	19.9
	5% WB	MCWBR	17.3	17.4	17.4	16.3	13.0	10.2	9.0	9.3	11.5	14.2	16.7	16.7
		MCWBR	19.5	22.1	24.7	25.4	23.3	20.2	19.1	18.2	18.7	20.3	19.7	17.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.305	0.324	0.358	0.387	0.429	0.487	0.487	0.481	0.396	0.361	0.325	0.305
		taud	2.440	2.367	2.357	2.312	2.225	2.099	2.112	2.116	2.327	2.435	2.496	2.495
	Ebn at Noon	Ebn at Noon	271	281	283	281	270	254	252	250	266	265	262	263
		Edh at Noon	25	31	35	40	44	50	49	48	36	29	24	22
(44) All-Sky Solar Radiation	RadAvg	567	853	1137	1479	1656	1802	1861	1610	1332	927	646	480	
	RadStd	48	97	90	145	191	155	98	103	168	102	61	54	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	-241	+266	+137	
(47) Regional (64 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+117	

Nomenclature: See separate page